F78PA Probability & Statistics A

Tutorial 1

1. Suppose that you throw three dice and that all outcomes of this experiment are equally-likely.
   (a) Describe the sample space Ω for this experiment.
   (b) Determine the probability that the numbers shown on the three dice are the same.
   (c) Determine the probability that the numbers on the three dice are all different.
   (d) Determine the probability that at least one of the dice shows an even number.

2. A lecturer gives her students a sheet with 9 probability problems and tells the class that the midterm test will consist of a random selection of 4 problems from the 9 problems on the sheet.
   (a) Determine how many different midterm question sets are possible.
   (b) Suppose that you know how to solve 6 of the 9 problems that were handed out. Determine the probability that
      i. you will be able to solve all 4 questions on the test.
      ii. you will be able to solve exactly 3 questions on the test.
      iii. you will be able to solve at least 2 questions on the test.

3. Suppose that $\Omega = \{1, 2, 3, ...\}$ is a countable sample space. (a) Show that the function $P(k) = \frac{3}{4^k}$ for $k = 1, 2, 3, ...$ is a probability function on the sample space Ω. (b) Let A be the event that the outcome of the experiment with sample space Ω is greater than 3. Compute $\mathbb{P}(A)$ without summing an infinite series.

4. Suppose that $\Omega = \{0, 1, 2, 3, ...\}$ is a countable sample space. Determine the value of c which makes the function $P(k) = c\frac{4^k}{k!}$ a probability measure for the sample space Ω.

5. The probability of the event A that construction work on a high rise building is cancelled because it is too wet is 0.35, and the probability of the event B that it is cancelled because it is too windy is 0.3. The probability (of the event) that it is simultaneously both too wet and too windy is 0.2. Find the probability that it is either too wet or too windy, and hence the probability that construction work can go ahead.

6. A physical device only works if the ambient temperature lies within the range $[T_1, T_2]$ (where $T_1 < T_2$). Suppose that the probability of the event A that the temperature is below T_2 is 0.6, and that the probability of the event B that the temperature is above T_1 is 0.75. Find the probability that the device works.

7. Suppose that $(\Omega, \mathcal{F}, P)$ is a probability model and $E, F \in \mathcal{F}$ are two events. Show that
   (a) $\mathbb{P}(E) = 1 - \mathbb{P}(E^c)$
   (b) $\mathbb{P}(E \cap F^c) = \mathbb{P}(E) - \mathbb{P}(E \cap F)$
   (c) $\mathbb{P}(E \cup F) = \mathbb{P}(E) + \mathbb{P}(F) - \mathbb{P}(E \cap F)$

8. A worker has asked her supervisor for a letter of recommendation for a new job. She estimated that the probability the she will get the job is 0.8 if her supervisor writes a strong reference, 0.4 if her supervisor writes a moderately strong reference, and 0.1 if her supervisor writes a weak reference. She further estimates that the probability that the supervisor writes a strong reference is 0.7, the probability that the supervisor writes a moderate reference is 0.2, and the probability that the supervisor write a weak reference is 0.1.

(a) Determine the probability that she will get the job.
(b) Given that she does get the job, determine the probabilities that she received:
    i. a strong reference from her supervisor;
    ii. a moderate reference from her supervisor;
    iii. a weak reference from her supervisor.
(c) Given that she does *not* get the job, determine the probabilities that she received: a strong reference from her supervisor; a moderate reference from her supervisor; a weak reference from her supervisor.

9. An actuarial trainee is planning to take three actuarial exams in the coming year. He plans to take the first exam in September. If he passes that exam, he will take the next one in May, and if he passes that one, he will take the third exam in the following September. If he fails any of the exams he will not be permitted to take the following exams within the next year. The probability that he will pass the first one is 0.9. If he passes the first one, then the conditional probability that he will pass the next one is 0.8, and if he passes both the first and the second exams, then the conditional probability that he will pass the third exam is 0.7.
    (a) Determine the probability that the trainee will pass all three exams.
    (b) Given that he did not pass all three exams, what is the conditional probability that he failed the first exam?

Exam Style Question

An experiment consists of tossing two fair 6-sided dice once.

(a) Determine the sample space Ω for this experiment. [1 mark]

(b) Specify the event that corresponds to outcomes in which the sum of the dots on the dice equals 7, and compute the probability of this event. [2 marks]

(c) Specify the event that corresponds to outcomes in which the sum of the dots on the dice is greater than 12 and compute the probability of this event. Would the probability of this event be different if the die was biased? [2 marks]

(d) State the definition of conditional probability. [1 mark]

(e) Determine the probability that exactly one dice shows a 6 given that the sum of the dots on the dice equals 7. [1 marks]